Chapter 3

Multiple Integrals

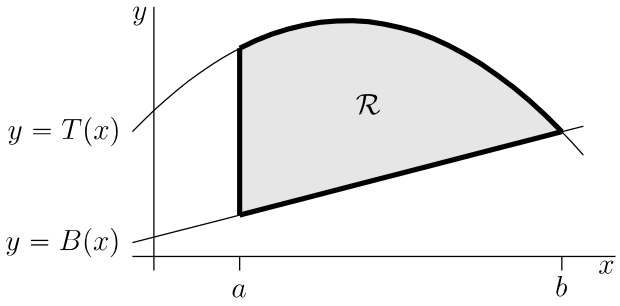
In your previous calculus courses you defined and worked with single variable integrals, like $\int_a^b f(x)\,\mathrm{d}x$. In this chapter, we define and work with multi-variable integrals, like $\iint_R f(x,y)\,\mathrm{d}x\,\mathrm{d}y$ and $\iiint_V f(x,y,z)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$. We start with two variable integrals.

3.1 Double Integrals

3.1.1 Vertical Slices

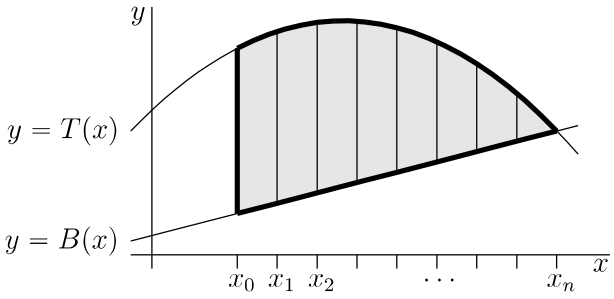
Suppose that you want to compute the mass of a plate that fills the region $\mathcal{R}$ in the xy-plane. Suppose further that the density of the plate, say in kilograms per square meter, depends on position. Call the density $f(x,y)$. For simplicity we'll assume that $\mathcal{R}$ is the region between the bottom curve $y = B(x)$ and the top curve $y = T(x)$ with x running from a to b. That is,

$$\mathcal{R} = \left\{ (x,y) \;\middle|\; a \le x \le b,\ B(x) \le y \le T(x) \right\}$$

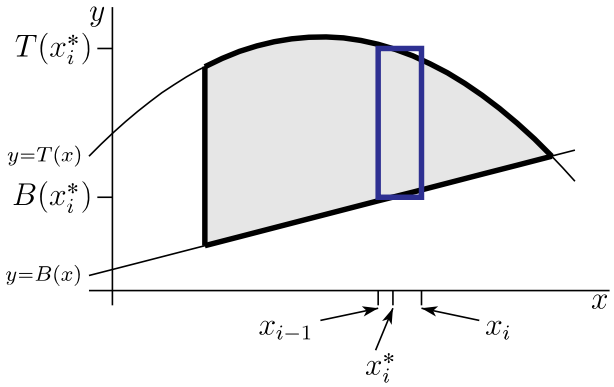


We'll shortly express that mass as a two dimensional integral. As a warmup, recall the procedure that we used to set up a (one dimensional) integral representing the area of $\mathcal{R}$ in Example 1.5.1 of the CLP-2 text.

- Pick a natural number n (that we will later send to infinity), and then
- subdivide $\mathcal{R}$ into n narrow vertical slices, each of width $\Delta x = \frac{b-a}{n}$. Denote by $x_i = a + i\,\Delta x$ the x-coordinate of the right hand edge of slice number i.



- For each $i = 1, 2, \ldots, n$, slice number i has x running from x_{i-1} to x_i. We approximate its area by the area of a rectangle. We pick a number x_i^* between x_{i-1} and x_i and approximate the slice by a rectangle whose top is at $y = T(x_i^*)$ and whose bottom is at $y = B(x_i^*)$. The rectangle is outlined in blue in the figure below.



- Thus the area of slice i is approximately $\big[T(x_i^*) - B(x_i^*)\big]\Delta x$.
- So the Riemann sum approximation of the area of $\mathcal{R}$ is

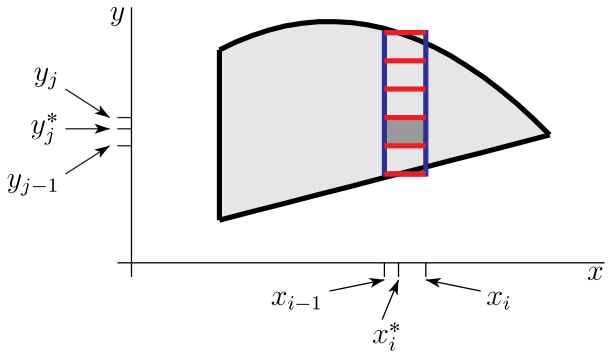
$$\text{Area} \approx \sum_{i=1}^{n} \big[T(x_i^*) - B(x_i^*)\big]\Delta x$$

- By taking the limit as $n \to \infty$ (i.e. taking the limit as the width of the rectangles goes to zero), we convert the Riemann sum into a definite integral (see Definition 1.1.9 in the CLP-2 text) and at the same time our approximation of the area becomes the exact area:

$$\text{Area} = \lim_{n\to\infty} \sum_{i=1}^{n} \big[T(x_i^*) - B(x_i^*)\big]\Delta x = \int_a^b \big[T(x) - B(x)\big]\mathrm{d}x$$

Now we can expand that procedure to yield the mass of $\mathcal{R}$ rather than the area of $\mathcal{R}$. We just have to replace our approximation $\big[T(x_i^*) - B(x_i^*)\big]\Delta x$ of the area of slice i by an approximation to the mass of slice i. To do so, we

- Pick a natural number m (that we will later send to infinity), and then
- subdivide slice number i into m tiny rectangles, each of width Δx and of height $\Delta y = \frac{1}{m}\big[T(x_i^*) - B(x_i^*)\big]$. Denote by $y_j = B(x_i^*) + j\,\Delta y$ the y-coordinate of the top of rectangle number j.



- At this point we approximate the density inside each rectangle by a constant. For each $j = 1, 2, \ldots, m$, rectangle number j has y running from y_{j-1} to y_j. We pick a number y_j^* between y_{j-1} and y_j and approximate the density on rectangle number j in slice number i by the constant $f(x_i^*, y_j^*)$.
- Thus the mass of rectangle number j in slice number i is approximately $f(x_i^*, y_j^*)\,\Delta x\,\Delta y$.
- So the Riemann sum approximation of the mass of slice number i is

$$\text{Mass of slice } i \approx \sum_{j=1}^{m} f(x_i^*, y_j^*)\,\Delta x\,\Delta y$$

  Note that the y_j^*'s depend on i and m.
- By taking the limit as $m \to \infty$ (i.e. taking the limit as the height of the rectangles goes to zero), we convert the Riemann sum into a definite integral:

$$\text{Mass of slice } i \approx \Delta x \int_{B(x_i^*)}^{T(x_i^*)} f(x_i^*, y)\,\mathrm{d}y = F(x_i^*)\,\Delta x$$

  where

$$F(x) = \int_{B(x)}^{T(x)} f(x, y)\,\mathrm{d}y$$

  Notice that, while we started with the density $f(x, y)$ being a function of both x and y, by taking the limit of this Riemann sum, we have "integrated out" the dependence on y. As a result, $F(x)$ is a function of x only, not of x and y.
- Finally taking the limit as $n \to \infty$ (i.e. taking the limit as the slice width goes to zero), we get

$$\text{Mass} = \lim_{n\to\infty} \sum_{i=1}^{n} \Delta x \int_{B(x_i^*)}^{T(x_i^*)} f(x_i^*, y)\,\mathrm{d}y = \lim_{n\to\infty} \sum_{i=1}^{n} F(x_i^*)\,\Delta x$$

  Now we are back in familiar 1-variable territory. The sum $\sum_{i=1}^{n} F(x_i^*)\,\Delta x$ is a Riemann sum approximation to the integral $\int_a^b F(x)\,\mathrm{d}x$. So

$$\text{Mass} = \int_a^b F(x)\,\mathrm{d}x = \int_a^b \left[\int_{B(x)}^{T(x)} f(x, y)\,\mathrm{d}y \right] \mathrm{d}x$$

This is our first double integral. There are a couple of different standard notations for this integral.